A Dashboard for Optimizing Prompts in Large Language Models

Angelo Broere, Luuk Versteeg, Taiki Papandreou-Lazos, and Tobie Werner



Fig. 1: Our prompt engineering component with each section denoted by a specific number: 1. Users can select a data sample to test. Information on the labels associated with the selected data sample is provided on the right. 2. This section allows users to create prompts by substituting values for each variable defined in the template. Users can customize the prompts to fit their specific needs and experimental setups. 3. The "Generate Prompts" button creates all possible combinations of prompts based on the user-defined variables. The "Test Prompts" button evaluates these generated prompts against the selected data sample, providing immediate feedback on their performance with colors. 4. This section visualizes the attention scores for each prompt. clicking word in the answer section lets you see distribution of the attention score in prompt.

1 Introduction

In recent years, large language models (LLMs) have gained substantial popularity due to their exceptional capabilities in comprehending and generating human-like text. Models such as GPT-3 and BERT, have revolutionized various natural language processing (NLP) tasks, from machine translation to text summarization. This surge has led to the emergence of prompt engineering, a technique widely adopted by a broad spectrum of users to fine-tune the performance of LLMs across various tasks. However, effective prompt engineering is often challenging, requiring a deep understanding of the model and the dataset. Consequently, creating intuitive interfaces that simplify and provide more insights into the prompt engineering process has become

increasingly relevant.
Our work specifically addresses the needs of prompt developers, particularly those who may not possess extensive expertise in their dataset fields, in optimizing prompts for text classification tasks, such as categorizing news articles. To this end, we have developed a comprehensive dashboard designed to visualize the effects of prompts on LLMs. This system provides an overview of datasets, suggests optimal prompt formats, and highlights important prompt components using attention scores. Inspired by PromptIDE [7], our dashboard emphasizes interaction and visualization to enhance user engagement and comprehension. The main contributions of this paper are summarized as follows:

- We provide detailed visualizations applicable to text classification datasets, aiding users in understanding datasets effectively.
- We introduce a novel approach to suggest optimally formatted and worded prompts, improving accuracy in text classification tasks.
- We offer interactive insights into the LLM/Transformer attention mechanism between the prompt and the result, improving the prompt engineering process.

By offering these features, our dashboard closes the gap between complex LLM capabilities and user-friendly interfaces.
Our code can be found on GitHub [1].

2 Related Work

The field of visual analytics for machine learning (ML) has seen substantial growth over the past decade, driven by the need to interpret complex machine learning models effectively. In this section, we review the most relevant references that frame our work in the current scientific context.
Wang et al. [8] provide a comprehensive survey of visual analytics for ML (VIS4ML), focusing on data-centric tasks across different stages of the ML pipeline. Their work is instrumental in offering a general introduction to the field, summarizing the impact of data quality on ML model performance, highlighting future research directions. Similarly, Law et al. [5] explore the perception of data insights among professional visualization users, providing a practical foundation for creating tools that effectively communicate data insights. These studies lay the groundwork for our exploration of visualization in data and model performance insights but lack specific techniques for prompt engineering and interactive optimization.
Prompt engineering has become pivotal in adapting LLMs for specific tasks. Strobelt et al. [7] introduced PromptIDE, a tool that allows users to experiment with and optimize language prompts. Their work illustrates the importance of interactive visual tools in enhancing prompt engineering. Mishra et al. [6] extend this idea through PromptAid, which assists non-experts in refining prompts without deep domain knowledge. Feng et al. [2] further this line of research with PromptMagician, focusing on efficient prompt recommendations and visual refinement techniques for text-to-image models. While these studies collectively inspire aspects of our approach, they do not fully integrate visual analytics with prompt engineering for LLM-based document classification, which is the focus of our work.
The exploration of complex data structures benefits greatly from innovative visual analytics techniques. Fischer et al. [3] propose a HyperMatrix for exploring temporal hypergraph models, emphasizing matrix-based visualization. This approach has influenced our integration of visual analytics, particularly in our dashboard's use of matrix-based visualizations to help users understand and refine prompt outputs effectively.
Understanding attention mechanisms in language models is crucial for improving model interpretability. DeRose et al. [1] and Wang et al. [9] contribute visual analytics tools—Attention Flows and Dodrio, respectively—that help users investigate the intricate attention patterns in transformer-based language models. Yeh et al. [11] add another layer with AttentionViz, which visualizes a joint embedding of query and key vectors to analyze global patterns across multiple sequences. Jaunet et al. [4] present VisQA, a tool designed to expose reasoning versus bias exploitation in visual question-answering models. These works have informed our methodology for visualizing attention distributions, yet they lack the integration part of these insights into an interactive prompt engineering framework. This is addressed in this dashboard interface.
Our interactive dashboard uniquely combines advanced visual analytics with prompt engineering specifically tailored for LLM-based document classification. By building on the foundational works cited, our tool offers an integrated platform that enhances the usability, efficiency, and effectiveness of prompt engineering. Our approach addresses the gaps in previous studies by providing a cohesive system that merges visual analytics with prompt optimization, thereby improving the overall user experience.

3 Task Definition

Problem Statement: The primary goal of our solution is to help prompt developers get insights into their datasets and experiment with various prompt configurations to enhance performance on text classification. The task involves classifying articles from an annotated dataset, where the objective is to predict the correct label for each article based on the given text and prompt. Our dashboard facilitates the exploration and optimization of prompts, thereby improving the accuracy and reliability of the LLM.
Target Audience: Our intended users are prompt developers with a fundamental understanding of ML but limited domain-specific knowledge. For instance, a developer skilled in general ML may not be versed in legal terminology. Our dashboard bridges this gap by providing intuitive visualizations and interactive tools, making prompt engineering accessible to users with diverse domain expertise.
Solution Overview: Our dashboard offers a user-friendly interface to support prompt developers through:

- **Dataset Visualization:** Visual tools for exploring label distributions and word frequencies.
- **Prompt Experimentation:** Interactive section for creating and testing different prompt formats to evaluate their result.
- **Attention Scores:** Visualization of attention scores to provide insights into the model's interpretation of prompts.
- **Prompt Performance Visualization:** Tools to compare prompt effectiveness using accuracy presented in a confusion matrix.

This comprehensive approach ensures that prompt developers can efficiently optimize prompts and enhance the performance of their text classification models, regardless of their domain knowledge level.

4 Components of the Dashboard Solution

To address the objective, our dashboard comprises three parts: data selection, prompt engineering, and prompt evaluation. Each component plays a crucial role in providing a comprehensive and interactive experience for users, enabling them to optimize their prompts efficiently.

4.1 Data Selection

The first component of our dashboard is dedicated to data analytics. This component includes dataset selection, data processing, data visualization, and an overview of the chosen samples.
As mentioned earlier, we draw our inspiration from promptIDE [7]. Consequently, we utilize the same datasets. These datasets include AG-news[2], Amazon Polarity[3], and GLUE[4].
The data selection interface, illustrated in Figure 2, allows users to select from the available datasets and choose specific splits, such as train, test, or validation. Upon selection, the dashboard provides a detailed description of both the dataset and the labels used to categorize each text.

[1] https://github.com/Luuk-Versteeg/MMA-group3
[2] https://huggingface.co/datasets/fancyzhx/ag_news
[3] https://huggingface.co/datasets/fancyzhx/amazon_polarity
[4] https://huggingface.co/datasets/nyu-mll/glue



Fig. 2: Data Selection Page: In section 1, users can select the desired dataset, specify the split of the dataset, and determine the number of data samples to be drawn. Below the selection buttons, a general description of the chosen dataset is provided, along with the labels used within the dataset. Section 2 allows users to visualize the label distribution and a word cloud of the data samples. Users can switch between these visualizations by clicking on the corresponding tab. Section 3 displays individual data samples and other data samples in a tabular form which allows users to scroll through them. By clicking on a specific sample, users can inspect it in detail.

To manage computational complexity, we implemented an option that allows users to customize the dataset size by specifying the desired number of data samples. The system then randomly samples this user-defined number from the full dataset, ensuring consistency across all components of the dashboard. This approach reduces the heavy computational demands of the model and allows users to conduct preliminary experiments on smaller subsets before scaling up to larger datasets. The selected data samples are displayed at the bottom of the interface, with individual samples viewable in detail upon selection. Additionally, a preprocessing activation button is implemented, applying preprocessing steps to the selected dataset. Preprocessing eliminates irregular tokens or identifiers, such as *#34;*, *\n* or *
* which were present in each dataset.

Building upon the work of promptIDE [7], we implemented a feature that provides pop-up suggestions for synonyms for each word in the text. In the single data sample viewer, users can hover over words underlined with a blue dashed line to view up to five synonyms. This functionality was initially designed to offer alternative word suggestions for prompts created by the user in the subsequent component. However, to enhance the interactivity of our dashboard, this feature is applied on the data sample as well.

The synonyms are derived from WordNet [5], a semantically oriented dictionary of English comprising 155,287 words and 117,659 synonym sets logically related to each other. For each word, we randomly select five synonyms. This approach was chosen to avoid the significant computational and memory costs associated with retrieving contextually relevant synonyms. While this method does not always provide contextually appropriate synonyms, we found that it delivers satisfactory results without incurring additional computational complexity.

[5] https://wordnet.princeton.edu/

Lastly, based on the chosen data samples, two types of visualizations are generated: a histogram of the label distribution and a word cloud representation of the most common words in the selected text samples. The histogram displays the distribution of labels within the selected data samples, providing a clear overview of label prevalence. The word cloud highlights the most frequently occurring words, with word size proportional to frequency. We present the top five most common words for each label. This allows users to gain a general understanding of the prevalent words associated with each label. This insight can later aid in identifying words that may pose challenges for LLMs.

4.2 Prompt Engineering

After selecting a dataset in the previous window, users can create prompts in the prompt engineering tab, as illustrated in Figure 1.

The selected data sample is displayed in the upper left corner of the interface. Users can navigate through the data samples using the left and right arrows on the screen. This component of the dashboard allows users to test their defined prompts on the chosen data sample.

The user starts by writing a prompt template in the designated text field. The prompt template is used to generate prompts that are input to the LLM. The template can contain variables, which are defined between curly brackets (e.g., {var1}, {var2}, etc). The user can define a number of variables. One special {text} variable is used to specify where texts from the dataset's entries should be placed.

For each variable (except the {text} variable), users can create multiple variants, each containing an input text box. Different variants could be different ways of phrasing the same sentence or question. The text entered in these variants is substituted into the prompt template at user-defined positions when generating prompts. This process is done by pressing the "generate prompts" button. All possible combinations of each variable's variants are made. For instance, if two variables are defined, with the first containing two variants and the second containing three variants, a total of six prompts will be generated. Additionally, hovering over text with a dashed underline will display suggestions for synonyms that can be used to enhance the variety of the prompts by providing alternative word choices. This enriches the prompt engineering process.

For our language model, we opted for TinyLlama[6], specifically the TinyLlama-1.1B-Chat-v1.0 version, loaded via the Huggingface pipeline[7] [12]. TinyLlama, with its 1.1 billion parameters, is significantly smaller than the largest Llama model, which has 65 billion parameters. TinyLlama was chosen due to its compact size and efficient performance. This suited our needs for quick, accurate, and local language processing.

The generated prompts are input into the aforementioned language model and the model will produce a human-like output. To determine if the model classified the text correctly, we look for an exact match to the ground truth label in the output. In case more than one of the possible labels is present in the model's response (e.g., both *positive* and *negative*), we select the most frequently occurring label as the model's prediction. We have observed that model responses containing multiple labels in varying quantities are responses similar to "*When choosing between positive and negative, I would say this text is negative.*". From a sentence like that, we pick *negative* as the predicted label. When the model's response does not contain any labels or equal amounts of different labels, we classify the text as *unknown*.

To get an indication of the performance of the prompts, the user can run the prompts on the selected data sample. The prompts resulting in a correct classification will be marked with a green background, and all incorrect classifications will be marked red.

Finally, for an even deeper understanding of the LLM's decision-making process, we visualize the attention scores generated by the language model. Attention mechanisms in LLMs work by assigning different weights to different parts of the input text, indicating which words or phrases the model considers most relevant when generating its

[6] https://github.com/jzhang38/TinyLlama

[7] https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0



Fig. 3: Prompt evaluation section featuring confusion matrix and result table. Section 1 provides general information about the selected dataset, giving users an overview of the dataset's characteristics. Section 2 displays a confusion matrix generated from the results of the selected prompt in Section 3. The confusion matrix allows users to evaluate the performance of the prompt by showing the distribution of correct and incorrect classifications. In section 3, users can choose which prompt to inspect more closely. The selected prompt samples are highlighted in blue, facilitating easy identification. Section 4 offers a detailed view of the selected prompts. Users can inspect the prompt itself and view a bar indicating the number of samples that were correctly classified, providing a clear measure of the prompt's effectiveness.

output. Visual attention scores can also reveal insights into the model's behavior, helping users to comprehend how different variables and their variants impact the model's predictions. This visualization allows the user to optimize their prompt engineering efforts more effectively.

4.3 Prompt Evaluation

The final component allows the user to evaluate the generated prompts on a larger subset of the dataset. The dataset used for this evaluation is sampled from the data selection component. The primary goal of this component is to enable users to assess the generalizability of the prompts that were constructed and refined in the prompt engineering section. The system tests the prompts created in the previous section against a broader array of data samples, which are then compared to the ground truth labels. A confusion matrix is subsequently constructed to display the model's response accuracy using the selected prompt. This is shown in Figure 3.

In the middle, a results table is presented. This table provides an overview of the classification results for each prompt. The "Total Scores" column indicates the number of data samples correctly classified by the model using the corresponding prompt format in that row. The subsequent columns present a detailed score for each label. If the model does not know the answer, the predicted label is marked as "Unknown". Additionally, users can select specific prompt rows to assess them individually in the section below. The selected prompts are then evaluated in the confusion matrix.

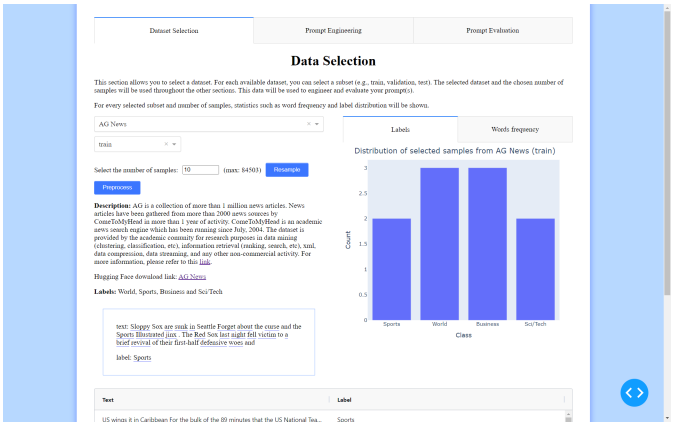


Fig. 4: The user selects the AG dataset in the dataset selection menu, specifically the training split. The application presents the user the dataset characteristics, a bar plot of label frequencies, and a word cloud. The user views a sample by clicking on it in the table.

5 Results

In this section, we present two use cases of our dashboard. The first use case covers the news classification task, and the second one covers Amazon's review binary sentiment classification task.

5.1 Use Case 1

The first use case takes AG-news dataset. Specifically, we extract 10 samples from the training split, with the label distribution presented in Figure 4. It is observed that the categories "World" and "Business" have a greater representation compared to "Sports" and "Sci/Tech". In addition, the user can inspect the word frequency visualized in a word cloud (see Figure 5. Only the 10 most common words are shown and the size indicates the number of occurrences. Figure 6 illustrates a prompt selected for evaluation in the prompt engineering component. This particular sample is chosen due to its inclusion of numerous indicative terms that characterize the sports genre. The format used for the prompts is var1 text var2, where var1 and var2 define the instructions, and the aforementioned data sample is inserted into text.

We define five different variants to {var1} and {var2} (see Figure 7), which forms a total of 25 combinations of prompts. For {var1}, we have the following values.

- Based on the content of the following news article:
- Given the following text:
- This is a news story:
- The following text needs to be labeled:
- The text is from a dataset with these possible labels World, Sports, Business or Sci/Tech.

Similarly, {var2} contains five unique values.

- Determine the most appropriate category: World, Sports, Business, and Sci/Tech
- Classify the text into one of the following categories: World, Sports, Business and Sci/Tech
- What is the topic of the text? Choose out of: World, Sports, Business and Sci/Tech
- To what class does it belong? Pick World, Sports, Business, or Sci/Tech
- Specify which label suits the text the most from the following options: World, Sports, Business or Sci/Tech



Fig. 5: Word frequency visualised in a word cloud

Labels: World, Sports, Business and Sci/Tech

text: Sloppy Sox are sunk in Seattle Forget about the curse and the Sports Illustrated jinx . The Red Sox last night fell victim to a brief revival of their first-half defensive woes and

label: Sports

Fig. 6: Selected data sample from the AG-news dataset.

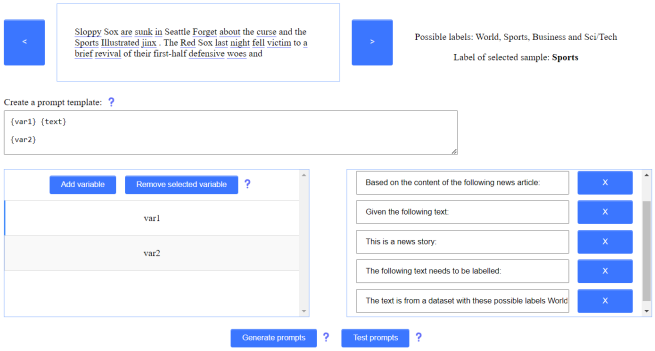


Fig. 7: Prompt engineering section where the user creates a prompt template and experiments with different variants

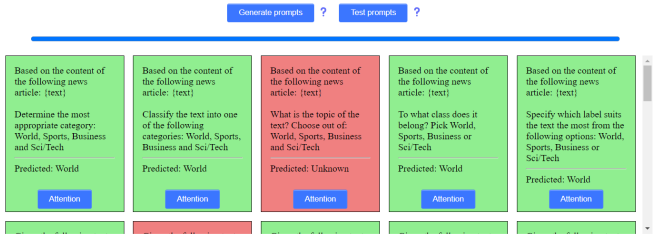


Fig. 8: Good prompts for the AG-news dataset

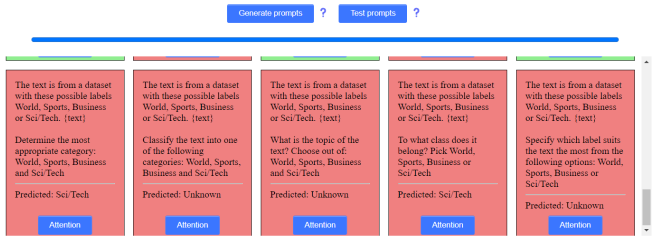


Fig. 9: Bad prompts for the AG-news dataset

These prompts along with the data sample text are tested. The result is shown in Figure 8. A prompt card is colored green if the prompt results in a correct model prediction, otherwise, it is red. We observe that, out of the five prompts containing the first variant of variable var1, only one causes the model to output an incorrect answer. On the other hand, the model predicts wrongly when provided with any of the five prompts containing the last variant of variable var1 (Figure 9). This indicates that using a prompt that contains the first variant is likely to be more effective when classifying this particular text sample.

After testing the prompts, we can gain an understanding of why the model produced a certain output. We do so by clicking the attention button of a tested prompt. This displays the prompt and the model's output in the attention visualizer, shown in Figure 10. When clicking on an output token in the attention visualizer we view the attention scores of the prompt's tokens. In this example, we see that the token "world" in the model's output was generated by primarily attending to tokens like "WASH" (from Washington), "passenger", "Islam", and "Security".

Lastly, to get more robust results on the accuracy of the prompts, we open up the prompt evaluation page. We press the "Run prompts" button and run all prompts using the 10 samples chosen in the initial dataset selection window. The window and its generated visualizations are shown in Figure 11. The table provides a comprehensive overview of the accuracy of the prompts. The best-performing prompt resulted in an accuracy of 4 out of 10 correct predictions. The accuracy in predicting individual classes is shown in separate columns. We then click on the rows of the table to populate the confusion matrix. In this example, we shift-click on all prompts to show the accumulated correct and incorrect predictions per class. Here, we see that the model usually predicts the label "World", indicating that the model has difficulty predicting the correct answer when given the previously constructed prompts.

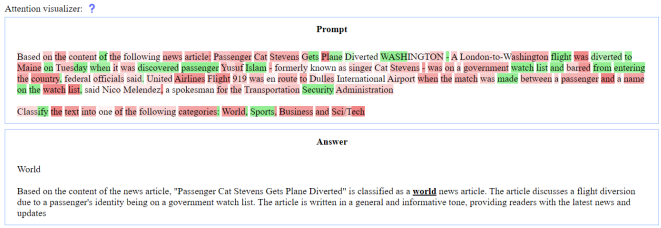


Fig. 10: Attention visualiser

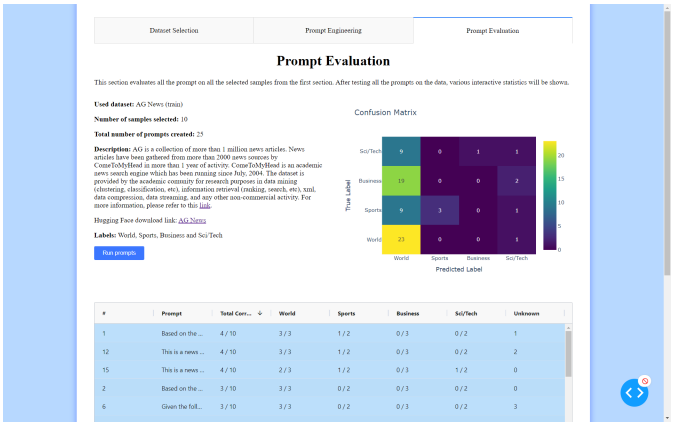


Fig. 11: Prompt evaluation

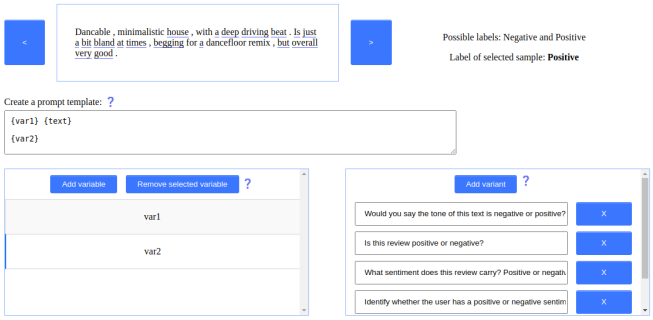


Fig. 12: Prompt engineering for the Amazon Polarity dataset

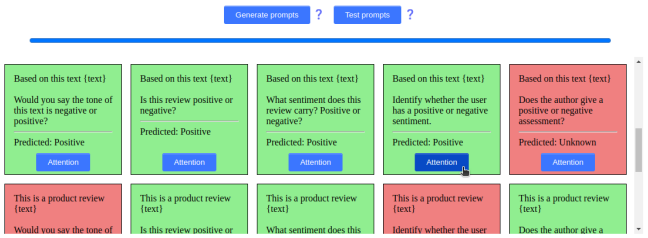


Fig. 13: Good prompts for Amazon Polarity

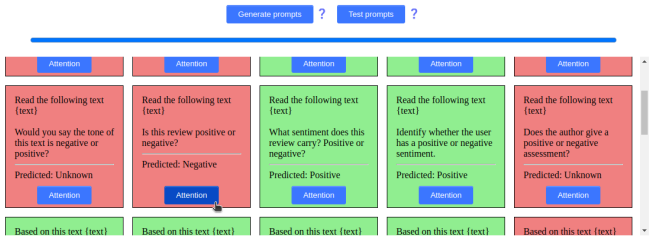


Fig. 14: Bad prompts for Amazon Polarity

5.2 Use Case 2

The second use case takes the Amazon Polarity dataset. In the *Data selection* tab (see Figure 2) we draw 25 samples from the *train* split. In the next tab, we define our prompt template and our variables. Our prompt template contain two variables, namely {var1} and {var2}. {var1} contains the following variants:

- Given this Amazon product review
- Read the following text
- Based on this text
- This is a product review
- Read the following text and classify it as positive or negative

And {var2} contains the following variants:

- Would you say the tone of this text is negative or positive?
- Is this review positive or negative?
- What sentiment does this review carry? Positive or negative?
- Identify whether the user has a positive or negative sentiment.
- Does the author give a positive or negative assessment?

The prompt template and the variants input is shown in Figure 12. Next, we generate all the prompts, i.e. creating all possible combinations of variants of variables within the constraints of the defined prompt template. These prompts can be tested on the selected sample above. After running the test, we can get an overview of which prompts perform well and which perform badly. We can see in Figure 14 that prompts starting with variation *read the following text* perform very poorly. Looking at Figure 13, we see that prompts starting with variation *Based on this text* classify the text correctly four out of five times.
The prompt in Figure 14 classified the text as *negative* while it is in fact *positive*. We can get an idea of why the model classified the text as *negative* by clicking *negative* in the Attention Visualizer. This shows us that the model gave a lot of attention to the negative word *bland* when generating the word *negative* in it's answer.
In the next and final tab we evaluate all of the prompts using all of the samples selected in the *data selection* tab. In Figure 15 we see that prompt four performs best, it classified 24 out of 25 text correctly. From our dashboard, we can derive that prompts starting with *Given this Amazon product review* classify the most texts correctly. The best-performing prompt is: *Given this Amazon product review: {text} Identify if the user has a positive or negative sentiment.* The way our dashboard is set up helps you identify what is a good prompt and what a prompt needs to perform well. In the aforementioned example, we see that it helps the LLM when you mention that the text is an Amazon product review, mentioning that the text is coming from a user helps the model even more.

6 Limitations & Future Work

There are several notable limitations in our dashboard. One significant limitation is in the synonym suggestion. Due to time constraints, we were unable to fully explore methods for providing users with optimal prompt formats. Our current implementation randomly suggests synonyms for each word using WordNet. This simplistic approach often yields irrelevant synonyms, as it lacks contextual understanding of the word in the text. In future work, we aim to enhance this feature by focusing on providing better phrasing rather than merely suggesting alternative words. Effective wording is more crucial than the specific words used in creating high-quality prompts, and our goal is to develop a more context-aware system to improve this aspect [10].
Another key limitation of our dashboard is the system's processing speed. Currently, the system operates on the Snellius GPU cluster, enabling us to evaluate each prompt variation in approximately four seconds. However, the running time increases linearly with the size of the dataset. This becomes particularly problematic when testing prompts on large datasets in the prompt evaluation section, as it can result in prolonged waiting times for generating results for each data sample across multiple prompt variations. Therefore, there is a significant need to improve the system and the inference speed of the language model to enhance overall efficiency.
Another limitation that needs to be addressed is the constraint in dataset and model selection. Currently, users are unable to incorporate additional datasets or different language models according to their preferences. The system is limited to selecting from three pre-defined datasets (although an extra dataset can be added manually in the source code)

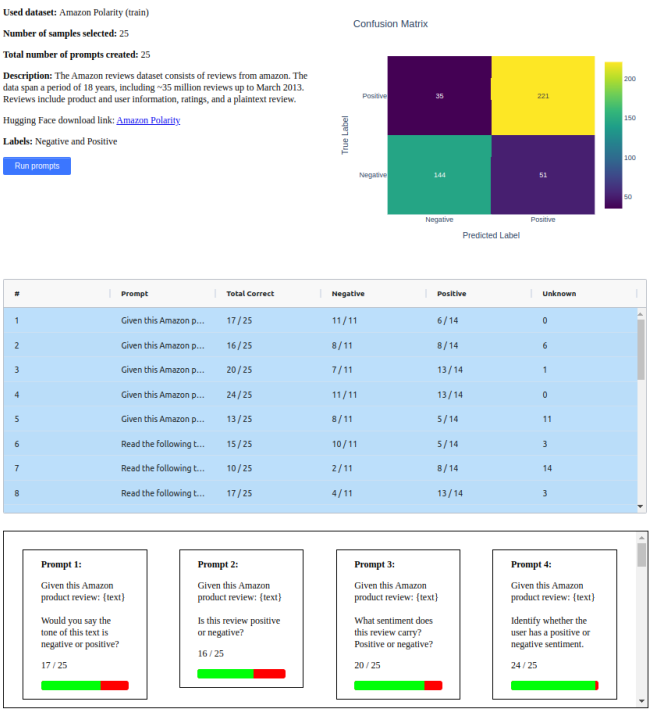


Fig. 15: Prompt evaluation for Amazon Polarity

and one language model. On top of that, we are currently randomly sampling from the dataset and this prohibits reproducibility of the result and also a more customized data sampling. For future work, we aim to implement an interactive method for incorporating additional datasets, providing more options for model selection and a more sophisticated way of data sampling. This enhancement will significantly increase the flexibility and usability of the dashboard.

Furthermore, the system currently lacks the ability to compare statistical properties between different datasets. This feature would enable users to gain deeper insights into various statistical properties, such as sentence length distribution and word variations across multiple datasets. For instance, users might want to examine differences in statistical properties between datasets within the same field. Implementing this feature would better address users' needs.

REFERENCES

[1] J. F. DeRose, J. Wang, and M. Berger. Attention flows: Analyzing and comparing attention mechanisms in language models. *IEEE Transactions on Visualization and Computer Graphics*, 27(2):1160–1170, 2020. 2

[2] Y. Feng, X. Wang, K. K. Wong, S. Wang, Y. Lu, M. Zhu, B. Wang, and W. Chen. Promptmagician: Interactive prompt engineering for text-to-image creation. *IEEE Transactions on Visualization and Computer Graphics*, 2023. 2

[3] M. T. Fischer, D. Arya, D. Streeb, D. Seebacher, D. A. Keim, and M. Worring. Visual analytics for temporal hypergraph model exploration. *IEEE Transactions on Visualization and Computer Graphics*, 27(2):550–560, 2020. 2

[4] T. Jaunet, C. Kervadec, R. Vuillemot, G. Antipov, M. Baccouche, and C. Wolf. Visqa: X-raying vision and language reasoning in transformers. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):976–986, 2021. 2

[5] P.-M. Law, A. Endert, and J. Stasko. What are data insights to professional visualization users? In *2020 IEEE visualization conference (VIS)*, pp. 181–185. IEEE, 2020. 2

[6] A. Mishra, U. Soni, A. Arunkumar, J. Huang, B. C. Kwon, and C. Bryan. Promptaid: Prompt exploration, perturbation, testing and iteration using visual analytics for large language models. *arXiv preprint arXiv:2304.01964*, 2023. 2

[7] H. Strobelt, A. Webson, V. Sanh, B. Hoover, J. Beyer, H. Pfister, and A. M. Rush. Interactive and visual prompt engineering for ad-hoc task adaptation with large language models. *IEEE transactions on visualization and computer graphics*, 29(1):1146–1156, 2022. 2, 3

[8] J. Wang, S. Liu, and W. Zhang. Visual analytics for machine learning: A data perspective survey. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 2

[9] Z. J. Wang, R. Turko, and D. H. Chau. Dodrio: Exploring transformer models with interactive visualization. *arXiv preprint arXiv:2103.14625*, 2021. 2

[10] J. White, Q. Fu, S. Hays, M. Sandborn, C. Olea, H. Gilbert, A. Elnashar, J. Spencer-Smith, and D. C. Schmidt. A prompt pattern catalog to enhance prompt engineering with chatgpt. *arXiv preprint arXiv:2302.11382*, 2023. 6

[11] C. Yeh, Y. Chen, A. Wu, C. Chen, F. Viégas, and M. Wattenberg. Attentionviz: A global view of transformer attention. *IEEE Transactions on Visualization and Computer Graphics*, 2023. 2

[12] P. Zhang, G. Zeng, T. Wang, and W. Lu. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*, 2024. 3